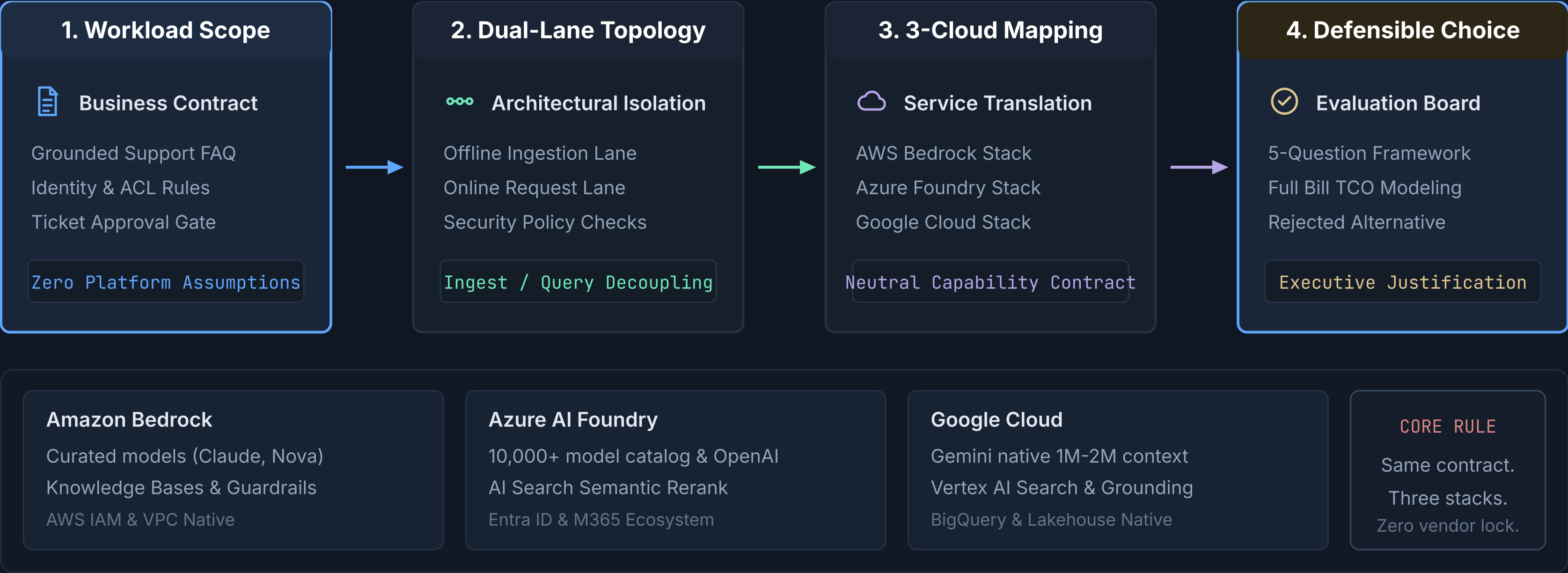


Cloud AI Capabilities Overview: Enterprise Architecture

Navigating AWS Bedrock, Azure AI Foundry, and Google Cloud for production GenAI.



Three Student Deliverables

What every engineering student must be able to design and defend by the end of this module.



Deliverable 1: Building Blocks

Identify Managed AI SaaS

Models, RAG, Agents, Evals, Traces, Governance

Mastery Verification Criteria:

- Deconstruct vendor catalogs into 6 capabilities
- Distinguish managed RAG from custom search
- Spot managed agent limits vs custom loops
- Identify the full 7-item production cost shape

Outcome: Capability Fluency



Deliverable 2: Cross-Cloud Map

1 Workload, 3 Realizations

AWS Bedrock, Azure Foundry, Google Cloud

Mastery Verification Criteria:

- Keep the identical business contract intact
- Map offline data ingestion across 3 clouds
- Map online query and human approval gates
- Enforce enterprise IAM & document ACLs

Outcome: Multi-Cloud Portability



Deliverable 3: Defend Choice

Architectural Defense

5-Question Framework & Trade-off Audit

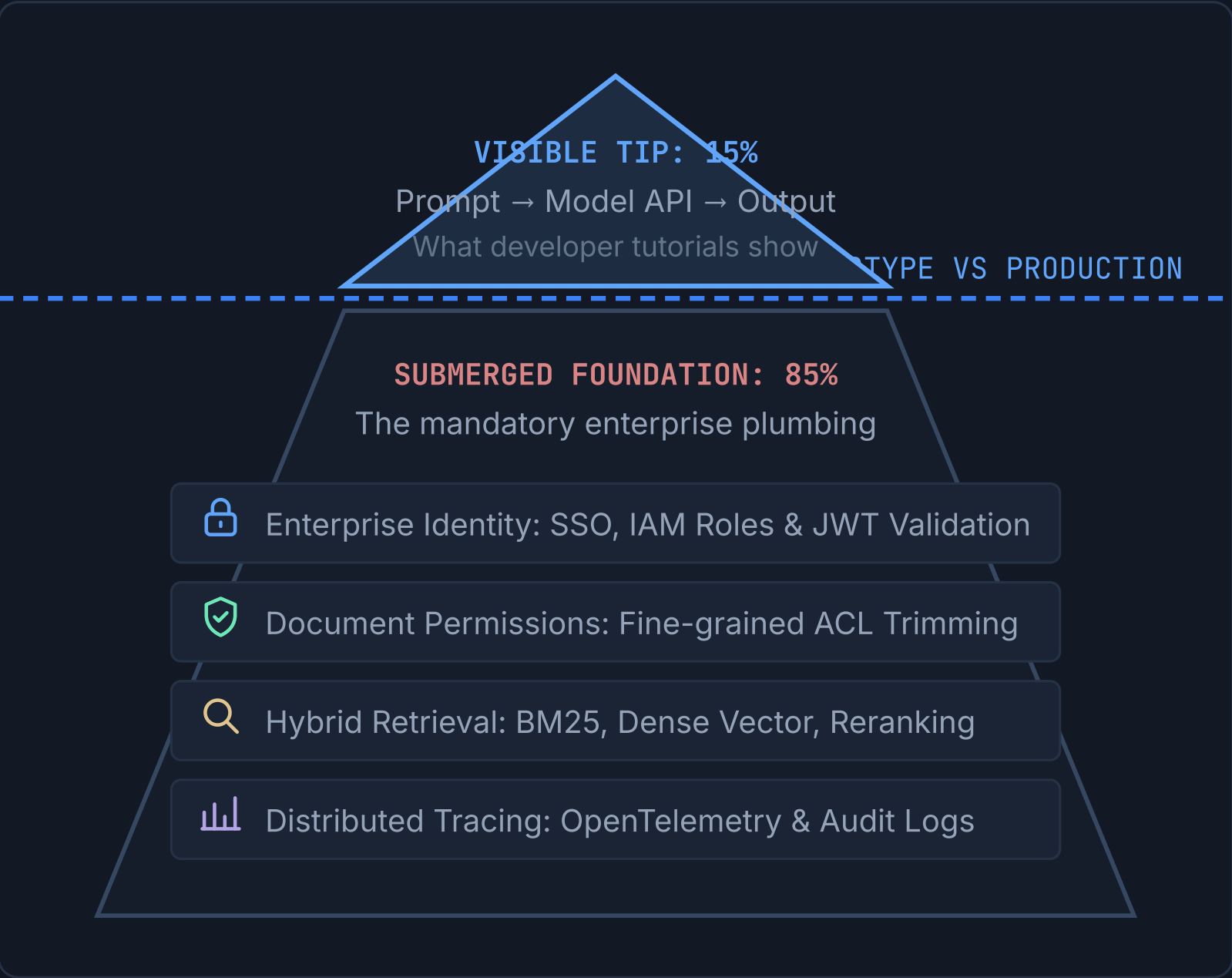
Mastery Verification Criteria:

- Evaluate data boundary and compliance rules
- Calculate cost drivers beyond raw token price
- State exactly which platform was rejected
- Defend against team operational constraints

Outcome: Executive Justification

Moving Beyond the Prototype: The Iceberg Problem

A raw model endpoint is only 15% of a production GenAI system.



Prototype Failures vs Production Reality

Failure Mode 1: Data Leaks & ACL Bypasses

Naive RAG indexes all corporate documents into a flat vector store. Result: Standard users retrieve restricted executive payroll records.

Failure Mode 2: Unbounded Latency & Cost

Zero semantic caching, uncompressed system prompts, and loops spinning autonomously on ambiguous questions inflate bills 10x.

Failure Mode 3: Silent Quality Degradation

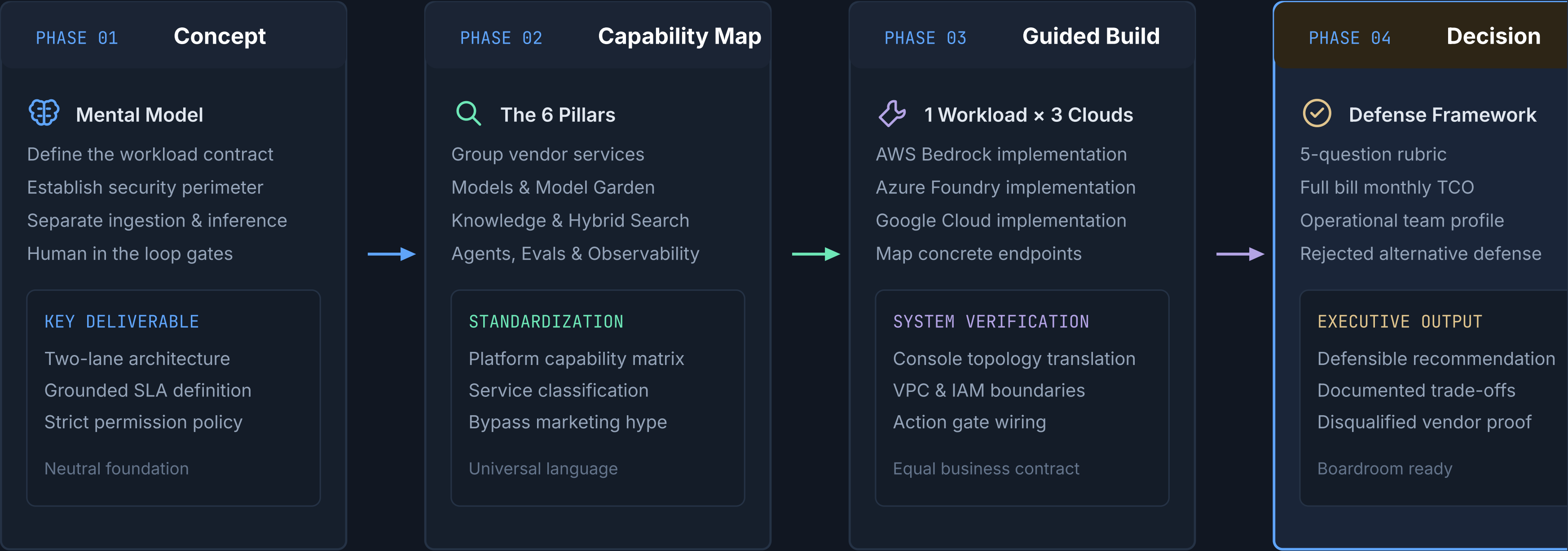
Document updates break table chunking: retrieval precision drops 30% with zero regression alerts or CI/CD test gates.

The Managed Cloud Advantage

Cloud platforms package IAM connectors, VPC isolation, vector indices, and automated evaluation so teams deploy securely in days, not quarters. Focus engineering on business logic, approval gates & golden evals.

The 4-Phase System Design Flow

Reusing official cloud architecture training methodology: Concept to Decision.



The 6 Pillars of Managed Enterprise AI

The standardized capability framework used to evaluate every cloud provider.

1. Foundation Models

Multi-provider catalog & open-weight weights
Serverless pay-per-token vs Provisioned Units
Adapter fine-tuning (LoRA) & distillation

AWS Bedrock • Foundry • Model Garden

2. Knowledge & RAG

Managed connectors (S3, SharePoint, Drive)
Automated chunking & embedding pipelines
Hybrid lexical + dense search & reranking

Bedrock KB • AI Search • Vertex Search

3. Agents & Orchestration

Stateful session memory & conversational state
Sandbox code interpretation & API invocation
Human in the loop approval confirmation gates

AgentCore • Foundry Agent • Gemini ADK

4. Evaluation & Benchmarks

Pre-release golden Q&A test sets
Automated Model-as-a-Judge scoring
Groundedness, relevance & toxicity metrics

Bedrock Evals • Foundry Evals • Vertex Eval

5. Tracing & Telemetry

Distributed spans via OpenTelemetry standard
Per-request token accounting & latency tracking
Failure clustering & error categorization

CloudWatch/X-Ray • App Insights • Trace

6. Enterprise Governance

IAM roles, SSO & document-level ACL filtering
Customer Managed Encryption Keys (CMEK)
Content safety, PII redaction & jailbreak filters

Guardrails • Content Safety • Model Armor

The Benchmark Case: Grounded Policy Assistant

A single real-world workload that exposes every critical enterprise boundary.

 USER QUERY (Sarah, Engineering):
"How long can I borrow a laptop, and where do I collect it?"
Context: Needs loaner equipment for client deployment

→

 GROUNDED ASSISTANT RESPONSE:
"Standard loans are up to 14 days. Collect at IT Helpdesk B2."
Citation: Equipment Loan Policy (2025 rev), Section 2.1

 Rule 1: Grounded Citation

No Evidence = No Answer

Must cite exact document name, section number

Pure parametric model knowledge is explicit

FAIL CONDITION

Hallucinated collection hours

 Rule 2: Identity & ACLs

Role-Based Trimming

Assistant can only search documents user has access to

Enforces Entra ID / IAM groups prior to LLM search

SECURITY CHECK

Pass JWT claims to vector filter

 Rule 3: Action Approval

Human Confirmation

Can draft a ServiceNow ticket, but cannot create it

Requires explicit user UI confirmation before action

APPROVAL GATE

Prevents unintended actions

 Rule 4: Audit & Trace

Zero Black-Box Runs

Full trace captured: prompt, retrieved chunks, tool calls

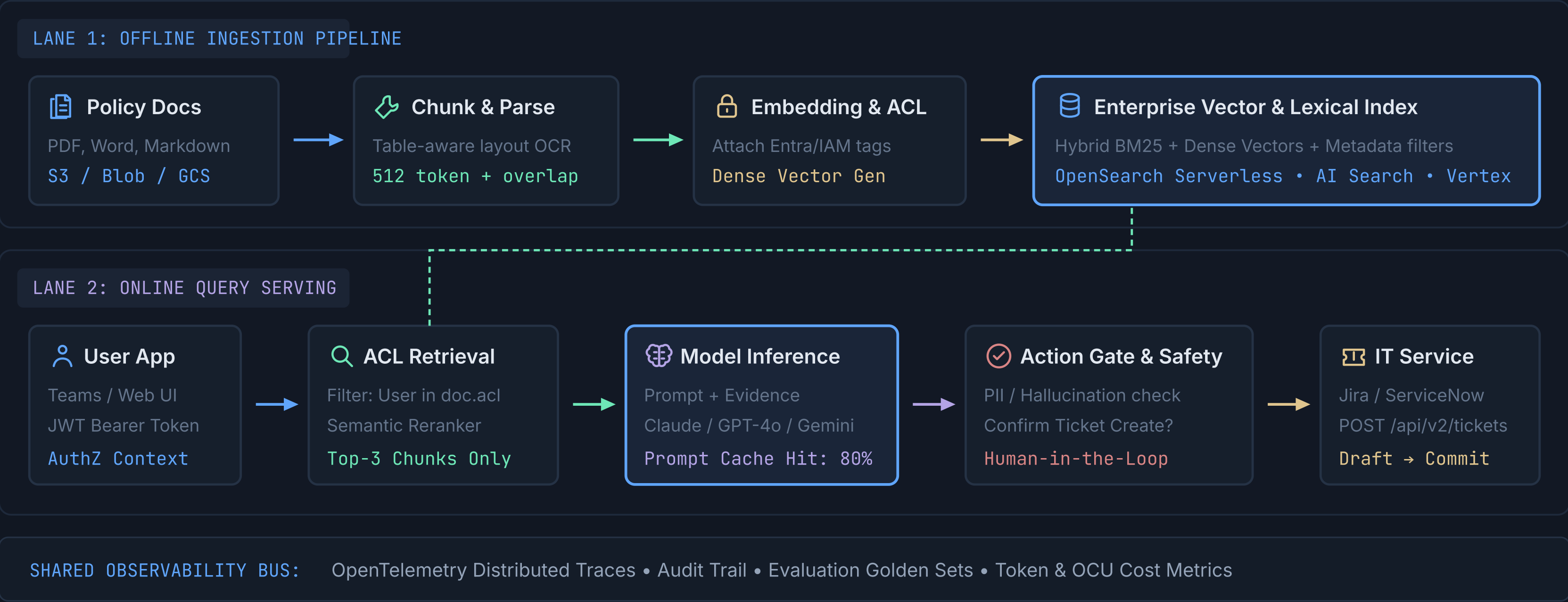
Logged to centralized observability for compliance

OTEL AUDIT LOG

Immutable compliance store

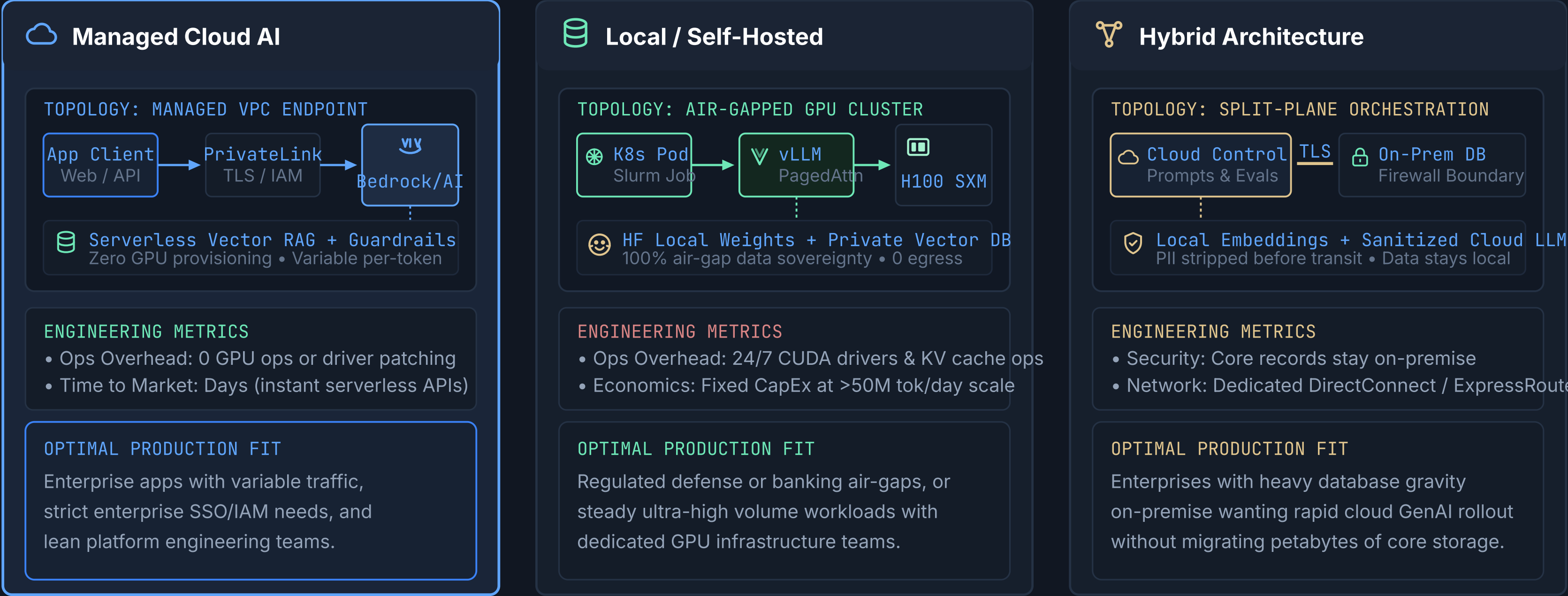
Dual-Lane Architecture: Offline Ingestion vs Online Request

Decoupling continuous document indexing from real-time low-latency inference.



The Tri-Model Deployment Spectrum

Selecting the operational model based on data classification, volume, and team maturity.



Model Catalog Philosophy: Curated vs Marketplace vs First-Party

Raw model counts are marketing signals: evaluate operational fit and regional availability.



Microsoft Azure AI Foundry

CATALOG PHILOSOPHY
Marketplace Hub • 10,000+ Models

 OpenAI: GPT-4o • o1 • o3-mini
Flagship Strategic Exclusive Partner

 Hugging Face: Direct Open Catalog

 Mistral Large

 Llama 3.3 70B

ENGINEERING EVALUATION

- API: Native OpenAI wire-format SDK
- SLAs: Tier-1 on OpenAI; OSS varies
- Privacy: Zero Data Retention via BAA
- Quota: PTU reservations for TPS SLA

OPTIMAL STACK FIT
OpenAI models & OSS marketplace



Amazon Bedrock

CATALOG PHILOSOPHY
Curated Multi-Provider Serverless

 Anthropic: Claude 3.5 Sonnet / Haiku
Flagship Reasoning & Coding Engine

 Amazon Nova Pro • Titan Embeddings V2

 Llama 3.3 70B

 Cohere Embed

ENGINEERING EVALUATION

- API: 100% Unified Converse API contract
- SLAs: Uniform AWS enterprise SLA
- Privacy: Zero Data Retention by default
- Quota: Serverless On-Demand + Provisioned

OPTIMAL STACK FIT
Claude 3.5 & unified Converse contract



Google Cloud Model Garden

CATALOG PHILOSOPHY
First-Party Multimodal & Open Models

 Gemini: 2.0 Flash • 1.5 Pro
1M to 2M Token Native Multimodal Context

 Claude 3.5 Sonnet + Gemma 2 (9B/27B)

 Llama 3.3 70B

 Mistral Large

ENGINEERING EVALUATION

- Context: Unmatched 1M-2M context tokens
- Multimodal: Audio, video, and PDF native
- TPU v5e: High-throughput Flash economics
- Caching: 75% savings via context caching

OPTIMAL STACK FIT
Native audio/video & 1M+ context

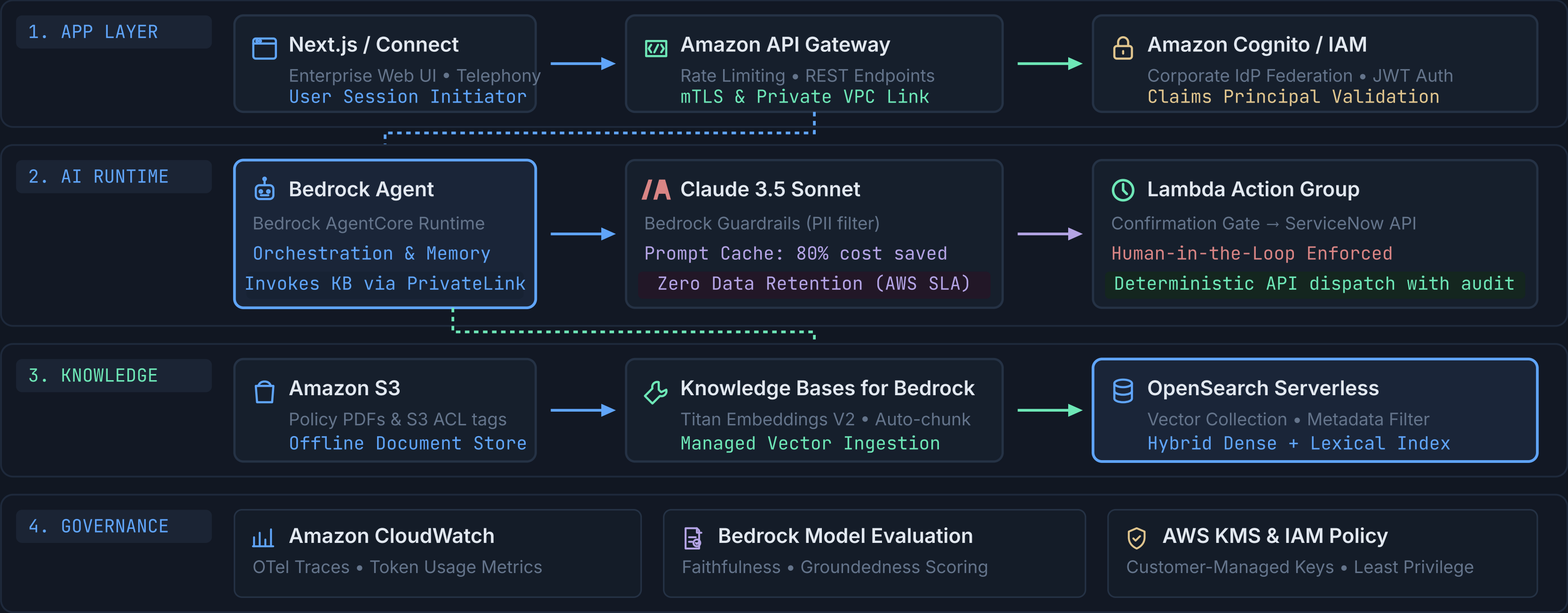
Qualitative Feature Matrix for the Support Assistant

Evaluating AWS, Azure, and Google Cloud across 8 enterprise dimensions.

ARCHITECTURAL DIMENSION	AWS BEDROCK	AZURE AI FOUNDRY	GOOGLE CLOUD
1. Frontier Model Access	Strong (Claude, Nova)	Differentiator (OpenAI)	Strong (Gemini 2.0)
2. Managed RAG & Reranking	Baseline (Bedrock KB)	Differentiator (AI Search)	Strong (Vertex Ground)
3. Agent Runtime & Tools	Strong (AgentCore)	Baseline (Foundry Agent)	Strong (Gemini ADK)
4. Automated Evaluations	Baseline (Bedrock Eval)	Strong (Foundry Evals)	Strong (Vertex GenAI)
5. Enterprise Identity & ACLs	Strong (IAM Roles)	Differentiator (Entra ID)	Baseline (Cloud ID)
6. Data Platform Co-location	Strong (S3 • OpenSearch)	Strong (Fabric • Cosmos)	Differentiator (BigQuery)
7. Organizational Gravity	Native for AWS Estates	Native for M365 Shops	Native for Data / GCP
8. Long-Context & Caching	Prompt Cache (Claude)	Prompt Cache (OpenAI)	Differentiator (2M Cache)

AWS Implementation: Bedrock-Centered Stack

How AWS services implement the two-lane support assistant topology.



When AWS Bedrock is the Strongest Choice

Strategic alignment, architectural advantages, and operational gotchas.

✔ Why Choose Amazon Bedrock?

1. Existing AWS Data Gravity

Enterprise policy documents, wikis, and databases already reside inside S3, RDS, or DynamoDB with mature IAM policies.

2. Enterprise Claude via Private VPC

Direct access to Anthropic Claude 3.5 Sonnet without public internet egress, routed through private AWS VPC endpoints.

3. Serverless Operational Model

Zero container clusters or server provisioning: Bedrock handles auto-scaling, cold-starts, and multi-AZ operational redundancy.

ORGANIZATIONAL ALIGNMENT

Strongest when the company is already an AWS shop: security, DevOps, IAM roles, and procurement pipelines are already in place.

⚠ Watch-Outs & Operational Gotchas

1. OpenSearch Serverless Cost Floor

Requires a minimum of 4 OCUs (2 search + 2 indexing), creating a fixed cost floor of ~\$700/mo even for low-traffic PoCs.

2. Knowledge Bases Chunking Rigidity

Standard chunking splits complex PDF tables poorly; custom chunking requires building an external Lambda preprocessing step.

3. Cross-Region Routing & Quotas

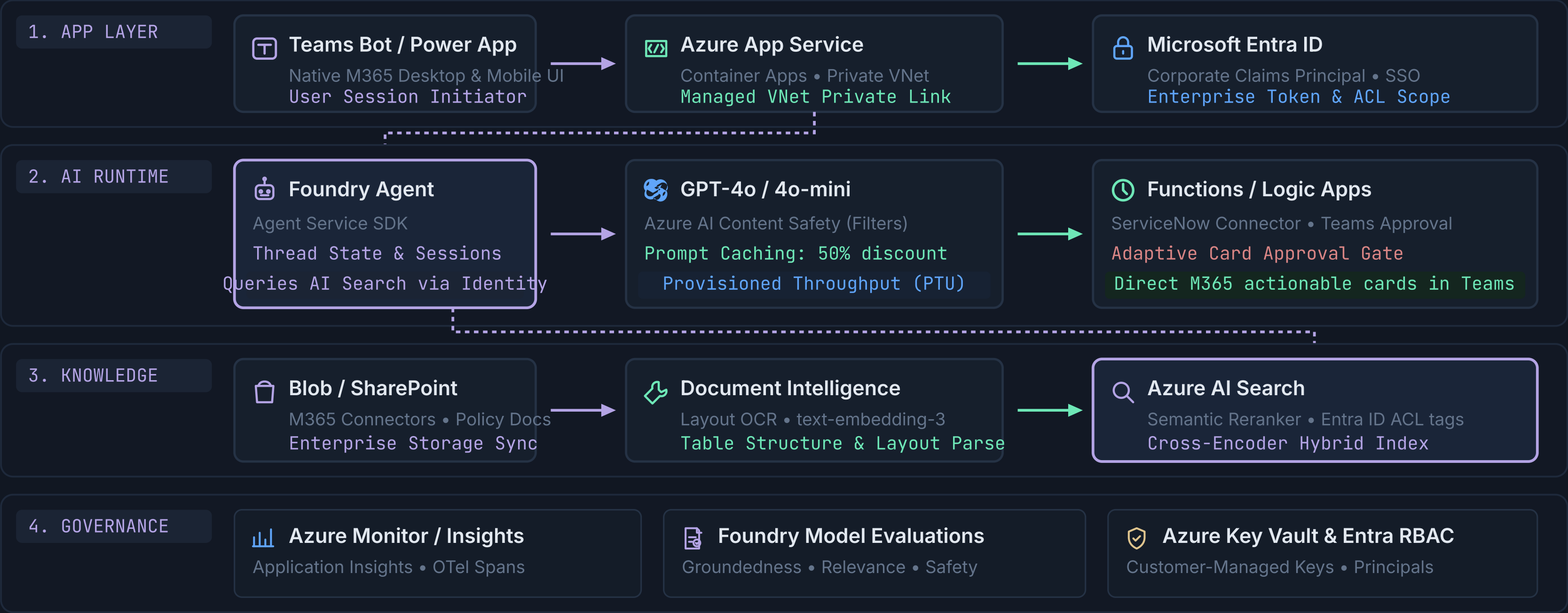
Claude 3.5 capacity in US-East-1 requires setting up cross-region inference profiles to avoid 429 Throttling exceptions.

ARCHITECTURAL MITIGATION

For pilot workloads under 10k queries/month, use Amazon Aurora PostgreSQL with pgvector instead of OpenSearch Serverless.

Azure Implementation: Foundry & AI Search Stack

Deploying the support assistant on the Microsoft enterprise ecosystem.



When Azure AI Foundry is the Strongest Choice

Corporate identity fit, M365 synergy, and architectural trade-offs.

✓ Why Choose Azure AI Foundry?

1. Native Microsoft Entra ID Trimming

Pass corporate JWT tokens directly into Azure AI Search: automatically enforces SharePoint document ACLs without custom sync scripts.

2. Enterprise OpenAI Models & SLAs

Priority access to GPT-4o, o1, and o3 with strict data guarantees: zero training on customer data and private virtual network endpoints.

3. Single Pane of Glass in Foundry

Unified portal consolidating model catalog, prompt playground, agent workflows, automated evaluations, and App Insights tracing.

ORGANIZATIONAL ALIGNMENT

Strongest for M365 enterprises: lowest friction path to launch grounded assistants inside Teams and Power Platform environments.

⚠ Watch-Outs & Operational Gotchas

1. Brand & Portal Renaming Churn

Fast product renaming (Azure OpenAI → AI Studio → AI Foundry) creates portal navigation churn and Python SDK version drift.

2. PTU Booking & Regional Quotas

Provisioned Throughput Units (PTUs) carry high monthly financial commitments: pay-per-token tier subject to strict TPM quota limits.

3. Azure AI Search Service Tier Costs

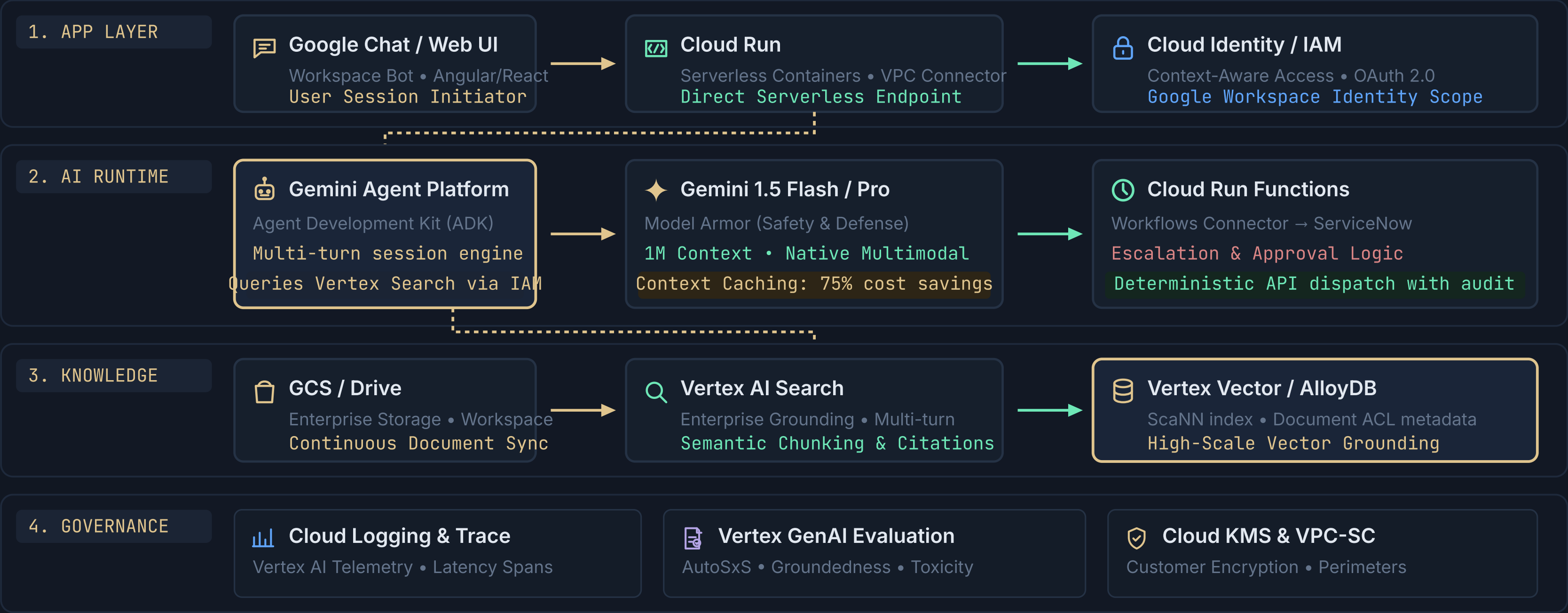
Semantic search features require Standard (S1) tier or higher, creating a ~\$250/mo baseline cost before query volume ramps up.

ARCHITECTURAL MITIGATION

Pin exact SDK versions (azure-ai-projects) and utilize Global Standard deployments to maximize throughput elasticity.

Google Cloud Implementation: Gemini Enterprise Stack

Orchestrating the assistant with Gemini Enterprise Agent Platform and Vertex AI.



When Google Cloud is the Strongest Choice

Analytics synergy, context length supremacy, and platform evolution.

✔ Why Choose Google Cloud?

1. BigQuery Lakehouse Co-location

If enterprise operational tables, customer service logs, and analytics live in BigQuery, Vertex AI provides zero-ETL grounding.

2. 1M-2M Native Context & Caching

Gemini native context windows allow entire equipment manuals or codebases to be cached in memory with up to 75% cost reduction.

3. Superior Flash Tier Unit Economics

Gemini 1.5 Flash delivers sub-second TTFT at pricing dramatically below comparable enterprise reasoning models.

ORGANIZATIONAL ALIGNMENT

Strongest for data-engineering and analytics-heavy firms that already run their primary data warehouse on Google Cloud Platform.

⚠ Watch-Outs & Operational Gotchas

1. Rapid Product Rebranding Pace

Transitions from Duet AI to Gemini Enterprise to Vertex Search create documentation churn and API parameter deprecations.

2. Regional Parity & Model Rollout

New Gemini checkpoints deploy first to us-central1; European and Asian regions often lag by several weeks or months.

3. Active Directory / Entra ID Friction

Federating legacy Windows Active Directory into Cloud Identity requires building and maintaining SAML/SCIM identity bridges.

ARCHITECTURAL MITIGATION

Deploy Workload Identity Federation for clean token exchange and explicitly lock model endpoint versions in Terraform.

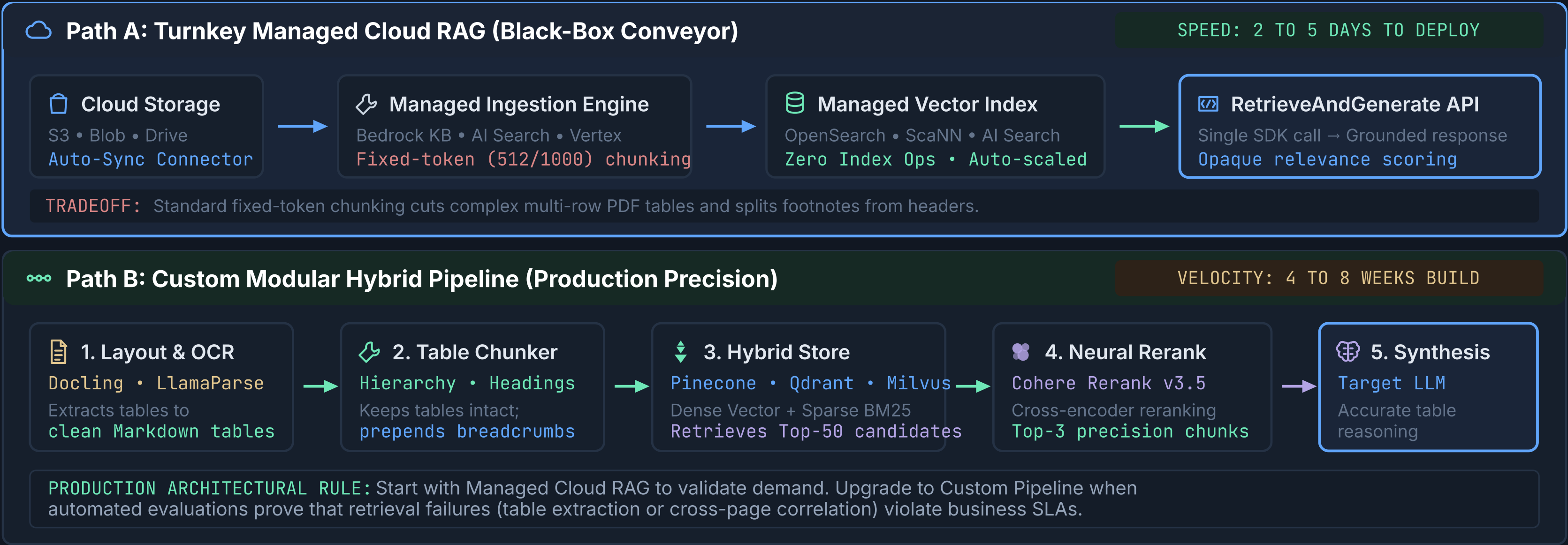
Side-by-Side Enterprise Component Alignment

Translating architectural functions across AWS, Azure, and Google Cloud.

ARCHITECTURAL FUNCTION	AMAZON BEDROCK	AZURE AI FOUNDRY	GOOGLE CLOUD (GEMINI / VERTEX)
Model Catalog & API	Bedrock Multi-provider • Converse	Foundry 10k+ • Azure OpenAI	Model Garden • Gemini 1.5/2.0 API
Managed Knowledge & RAG	Knowledge Bases for Bedrock	Azure AI Search (Semantic Rerank)	Vertex AI Search & Enterprise Grounding
Agent & Tool Orchestration	Bedrock Agents • Bedrock AgentCore	Azure AI Foundry Agent Service	Gemini Agent Platform • ADK
Content Safety & Guardrails	Bedrock Guardrails (PII & Prompt)	Azure AI Content Safety	Model Armor • Vertex Safety Filters
Tracing & Observability	Amazon CloudWatch • AWS X-Ray	Azure Application Insights • OTel	Google Cloud Observability (Trace)
Model Evaluation	Bedrock Model Evaluation	Foundry Evaluation Hub	Vertex AI Model Evaluation
Enterprise Identity & Auth	AWS IAM • Amazon Cognito	Microsoft Entra ID (Native M365)	Cloud Identity • IAM Workload Auth

Enterprise Knowledge: Managed RAG vs Custom Hybrid Pipeline

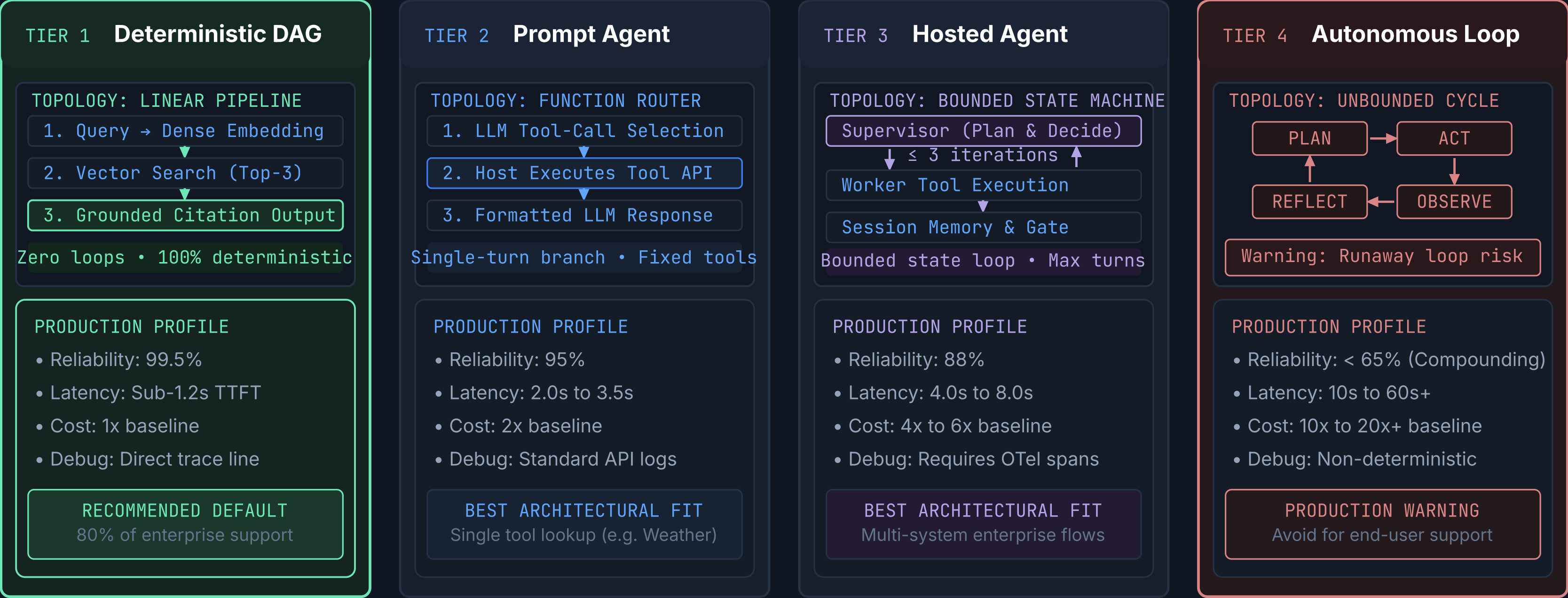
Balancing development velocity against parsing quality and retrieval precision.



PRODUCTION ARCHITECTURAL RULE: Start with Managed Cloud RAG to validate demand. Upgrade to Custom Pipeline when automated evaluations prove that retrieval failures (table extraction or cross-page correlation) violate business SLAs.

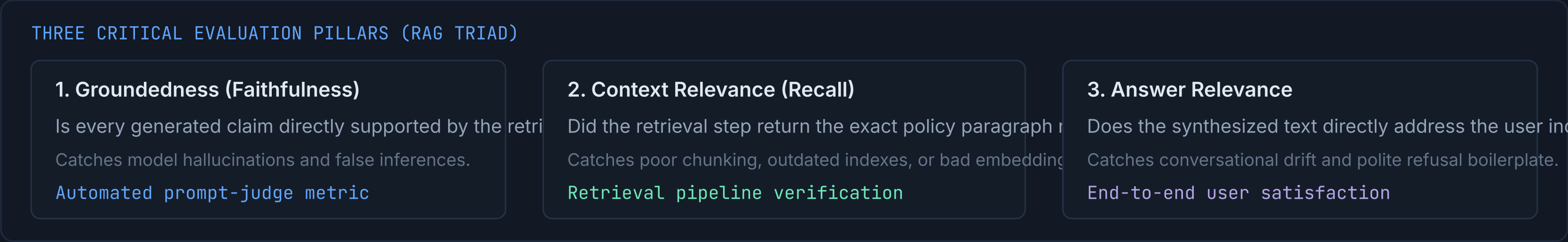
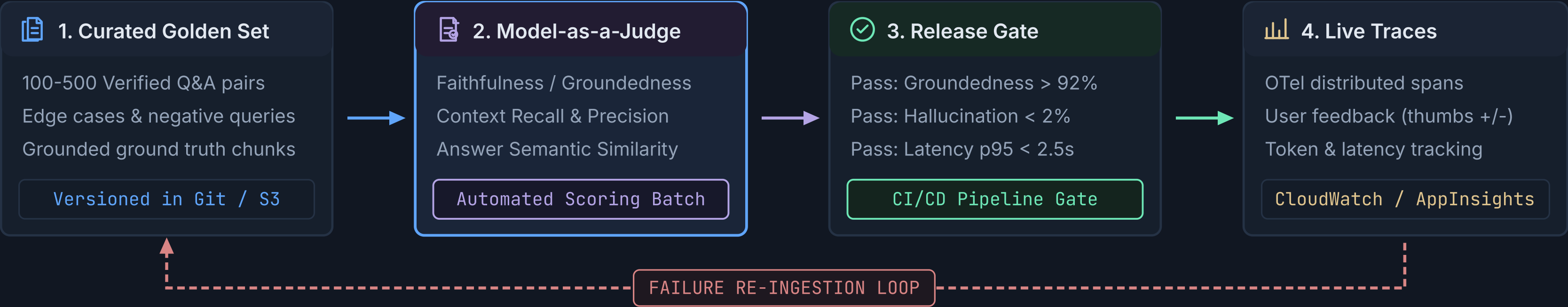
Agent Runtime Choices: When Do You Actually Need an Agent?

The spectrum of orchestration complexity: avoid over-engineering.



The Production Feedback Loop: Eval, Trace, & Mitigate

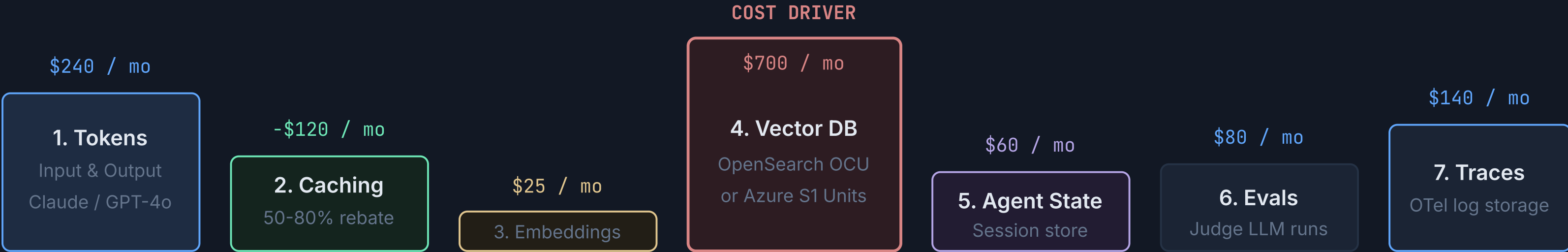
If you cannot trace a bad answer, you cannot operate an enterprise AI system.



The Anatomy of an Enterprise GenAI Bill

Raw token price is only one component: dissecting the 7 production cost categories.

MONTHLY PRODUCTION COST STACK (MODERATE TRAFFIC: 50,000 QUERIES/MO)



PRODUCTION COST REALITIES & LESSONS

- 1. Vector Minimums Dominate**

OpenSearch Serverless (4 OCUs) or Azure AI Search (S1) cost \$250-\$700/mo.

Mitigate: Aurora pgvector for small apps
- 2. Prompt Caching Saves 50-80%**

Static system instructions & policy chunks qualify for high-speed cache read rates.

Mitigate: Order prompt static-first
- 3. Tracing High-Frequency Overhead**

Logging full prompt payloads to CloudWatch rapidly accumulates gigabytes of ingestion.

Mitigate: 10% sampling on successful runs

Cost Optimization: Engineering Architectural Efficiency

Low cost is a design outcome, not a vendor discount: four engineering levers.



1. Context Compaction

30-50% Savings

Trim Boilerplate Before Prompt Assembly

- Strip repetitive HTML headers, legal disclaimers, and conversational fluff.
- Clean markdown extraction preserves tabular structure in half the tokens.

Architecture: Client-side AST pruning pipeline



2. Semantic & Prompt Caching

50-80% Savings

Bypass Model Inference on Repetitive Queries

- Exact match + cosine vector cache on Redis for common policy FAQs (0 tokens)
- Provider prompt caching (Bedrock Claude / Azure OpenAI / Gemini) on system prompts.

Architecture: Prefix caching with static system guidelines



3. Model Tiering & SLM Routing

70% Savings

Route by Workload Difficulty

- Send 80% simple lookup questions to SLMs (Haiku / 4o-mini / Flash).
- Escalate multi-step exceptions to Frontier models (Sonnet / GPT-4o / Pro).

Architecture: Lightweight intent classification gateway



4. Batch & Flex Pricing

50% Off-Peak Discount

Decouple Non-Interactive Inference

- Run offline document chunk summarization and embedding rebuilds overnight.
- Execute regression evaluation test batches via 24h batch APIs at 50% discount.

Architecture: Asynchronous SQS / Service Bus queues

The 5-Question Platform Selection Framework

The structured method for producing an executive-ready architectural recommendation.

Q1

Workload Pattern & Interaction Shape

Is this a deterministic policy FAQ, complex multimodal document parser, or autonomous multi-step agent?

Drives: Runtime Choice

Q2

Data Gravity, Residency & Document ACLs

Where do source documents currently reside (S3, SharePoint, BigQuery)? What are cross-border compliance boundaries?

Drives: Cloud Gravity

Q3

Model Capability Tier & Context Window

Does the application require 1M+ context (Gemini), enterprise Claude (Bedrock), or OpenAI reasoning models (Foundry)?

Drives: Model Selection

Q4

Enterprise Identity & Observability Baseline

Which identity provider (Entra ID vs AWS IAM) and tracing standard (OpenTelemetry) are approved by corporate SecOps?

Drives: Governance Fit

Q5

Operational Team Profile & Operator Burden

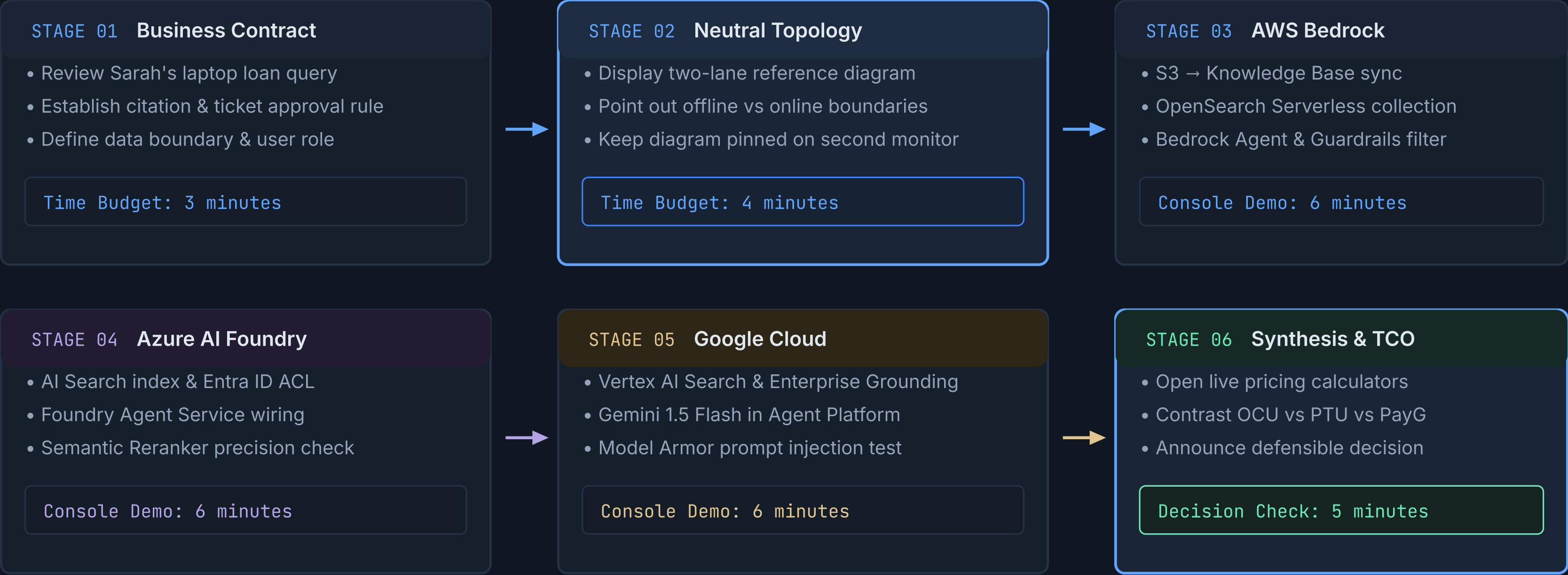
Can the team maintain custom Kubernetes clusters and vLLM GPU servers, or does the workload require serverless SaaS?

Drives: SaaS vs Self-Host

MANDATORY DEFENSE RULE: MUST EXPLICITLY NAME AND DISQUALIFY ONE REJECTED ALTERNATIVE

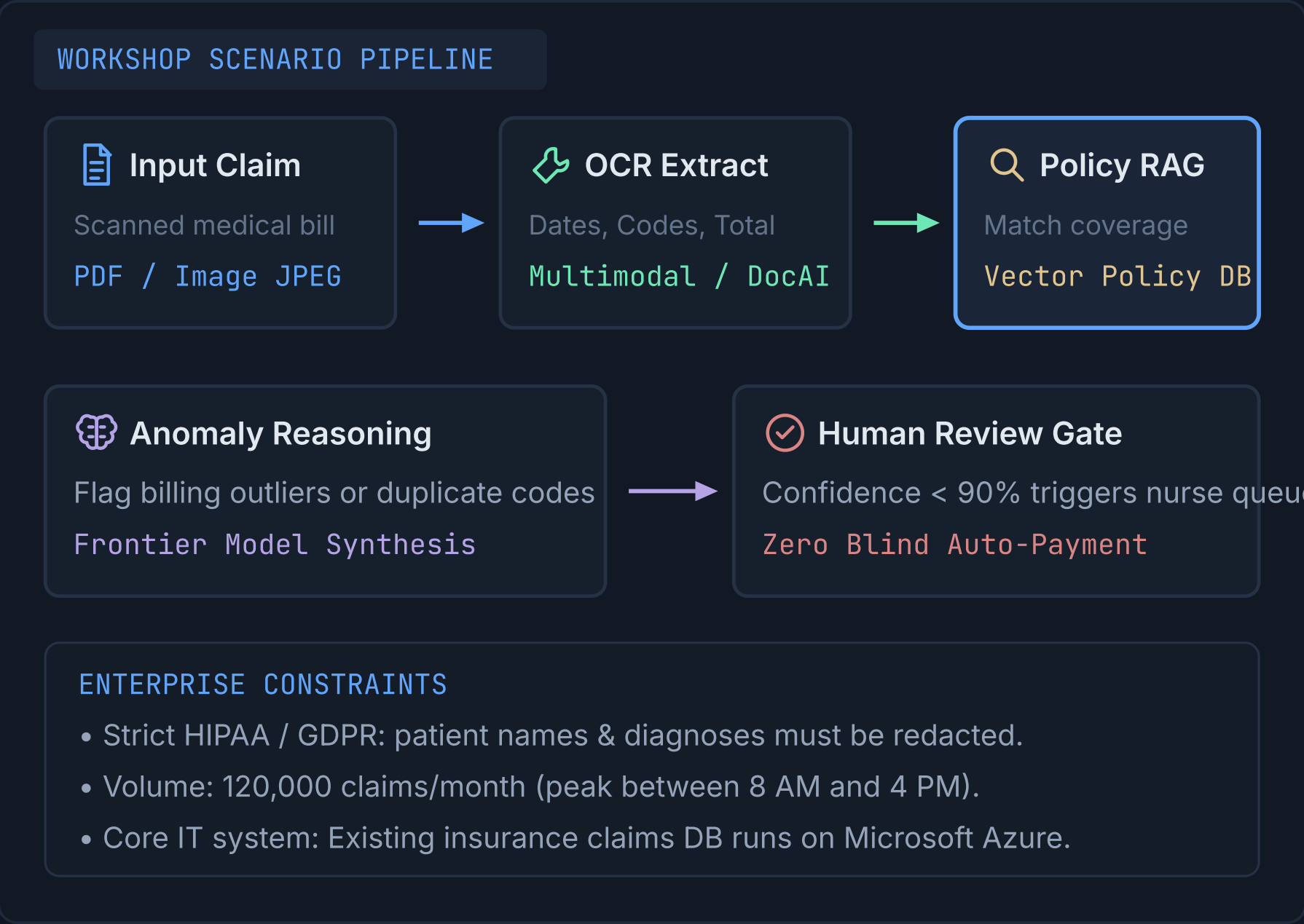
The Multi-Cloud Demonstration Flow

Demonstrating one architectural contract across AWS, Azure, and Google Cloud consoles.

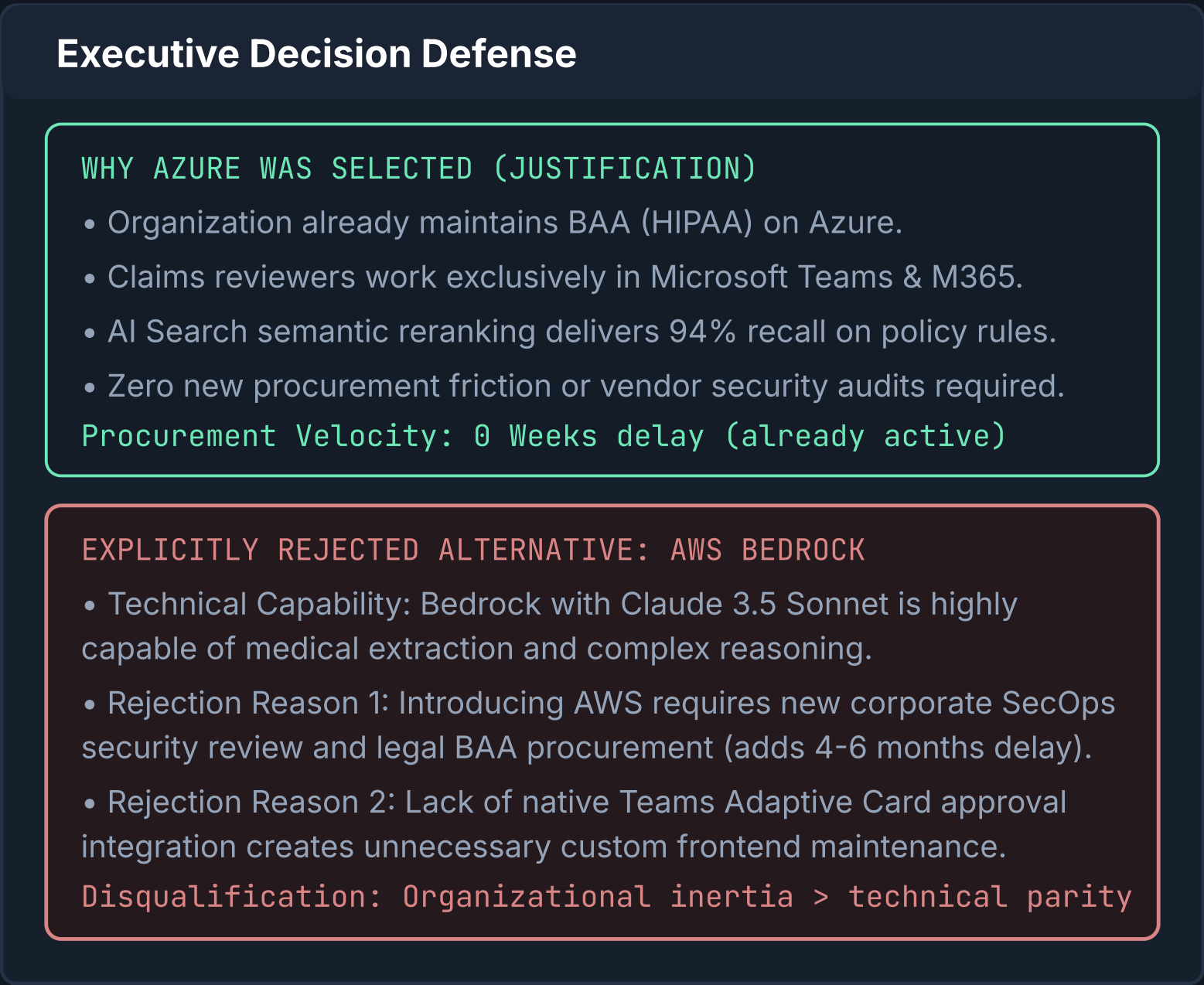


Design Task: Medical Claims Processing Assistant

Apply the 5-question framework to an end-to-end multimodal enterprise workload.



What an executive-ready architecture decision package looks like.



The Cloud AI Golden Rule

Managed platforms remove infrastructure work: they do not remove architecture decisions.

1. Evidence & Grounding

Where does the factual evidence live?
Never rely on model memory alone.

2. Identity & ACLs

Who is authorized to see each chunk?
Trim documents before prompt injection.



THE CLOUD AI GOLDEN RULE

Cloud platforms manage the models and hardware.
You own the boundaries, trust, and business risk.

3. Human Approval Gates

Which actions require human sign-off?
Zero blind ticket/payment execution.

4. Systematic Evaluation

How is quality continuously tested?
Golden sets and automated judges.

5. Observability & TCO

How are errors traced and bills tracked?
Vector minimums, OTel spans, tokens.